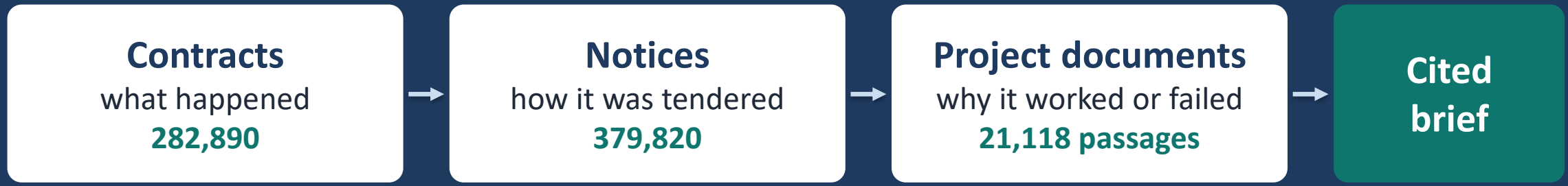
Procurement Intelligence Platform

Planning teams have contracts, notices and project-document lessons in separate places. This joins them into one cited planning brief.



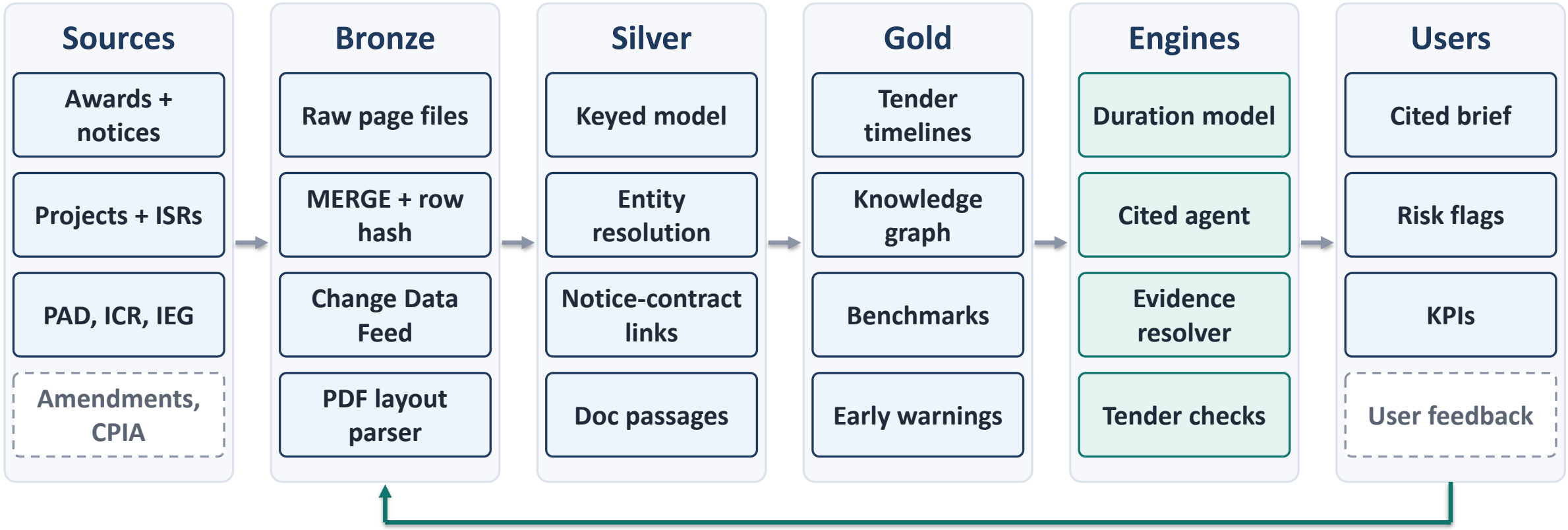
Shabad Vaswani | AI Engineer interview, World Bank | 27 October 2026

A working prototype on real World Bank open data, built on Azure Databricks

Architecture: one governed copy, from sources to users

Files, tables, graph, embeddings and models all live in the lakehouse under Unity Catalog

designed, not built

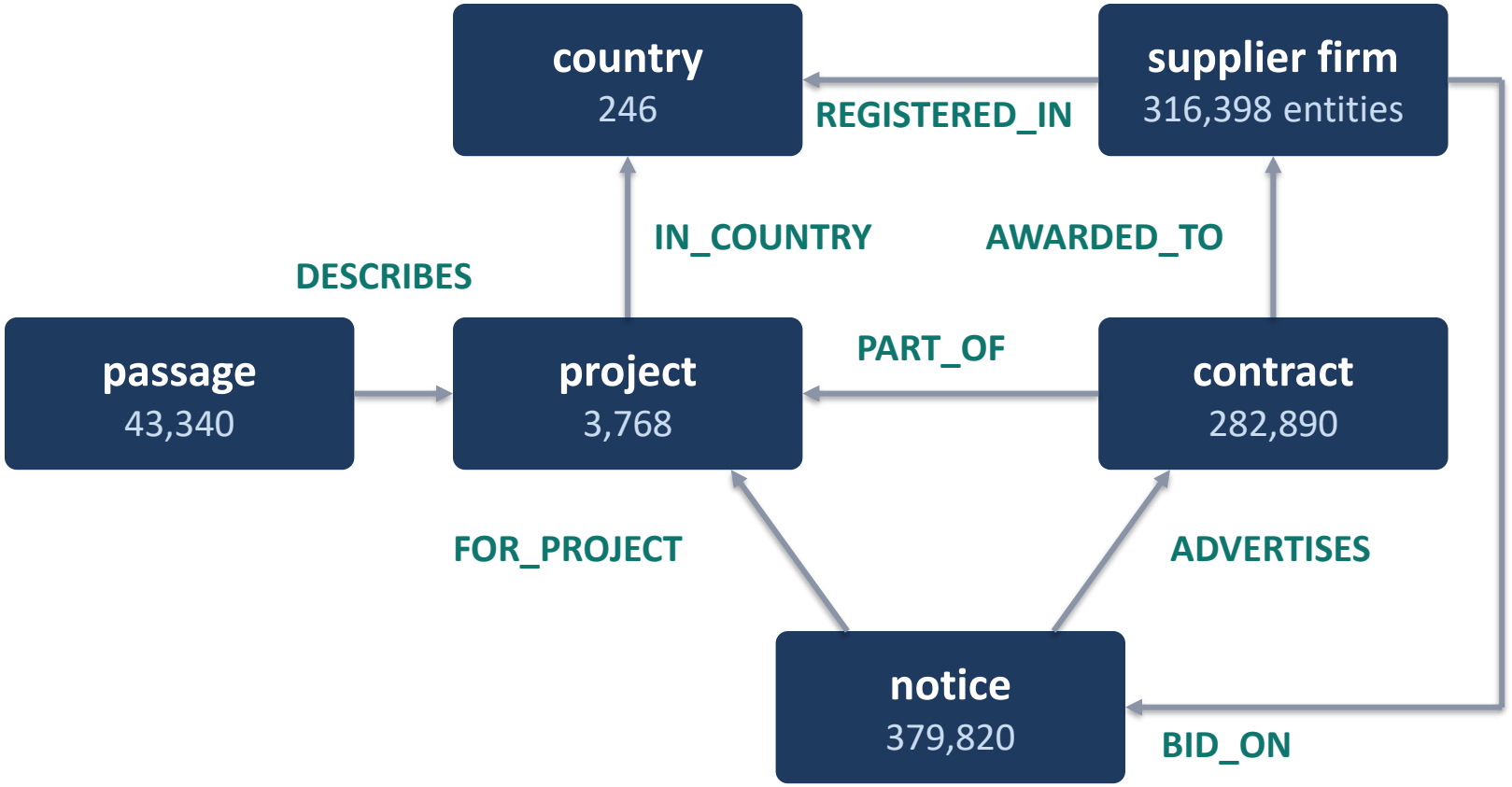


Daily loop: changed rows only, the graph updated by one MERGE, drift checks, champion/challenger gate

Unity Catalog | **MLflow** | **Workflows** | **Model Serving**

A keyed data model, and a knowledge graph built from it

Grain and keys first; every graph edge comes from a key that was checked



Delta node and edge tables; SQL joins and GraphFrames gave the same answer

0 orphans

and 0 duplicate keys on 25 checks; a breach stops the notebook

261,169 firms

from 344,273 supplier ids; precision 1.00, recall 0.90

2.28M edges

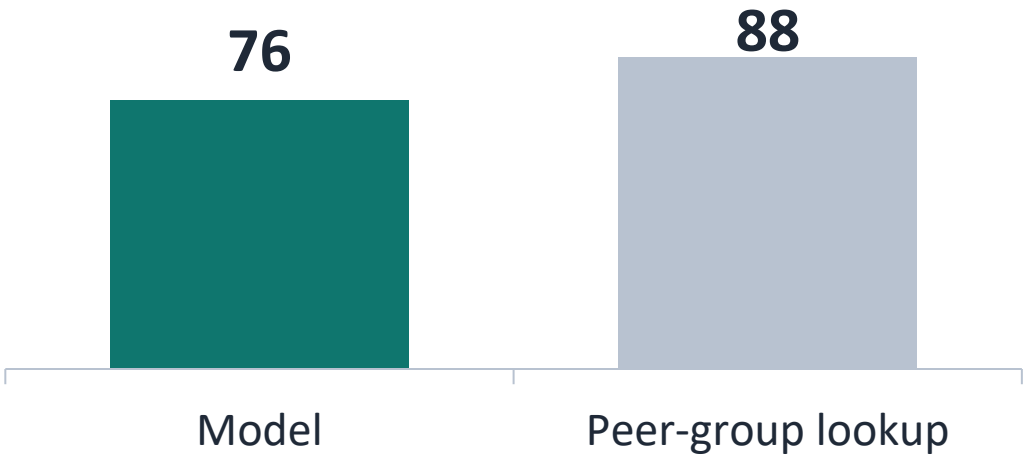
1.03M nodes, 0 dangling edges

Engines: a calibrated forecast and an agent that must cite

Statistics come from models and tables; the LLM plans, retrieves and writes, and code checks it

Duration and slow-tender risk

P50 error in days, 6,624 test tenders



80.7%
P10-P90 coverage, after a conformal step

0.747
ROC-AUC for slow tenders; peer rate 0.584

Cited agent

Input guard Llama 3.3 checks scope and injection

↓

Planner + 5 tools Qwen 3.5, at most 6 tool calls

↓

Writer JSON brief, a record id on every claim

↓

Validation in code unknown ids dropped, facts checked

↓

Cited brief timeline, risks, precedents, evidence gaps

The model is never the last check.

Evaluation and trust

Tested as of the notice date, with every label's origin stated

87 vs 100 days

timeline error on 300 real tenders replayed as of their notice date: model vs peer lookup

5 → 2 of 30

briefs with an unsupported claim, before and after facts lines and a code fact-check; 1 of 10 on a holdout

35 of 68

projects IEG rated unsatisfactory still had a satisfactory last ISR; both sides are shown

How the evaluation labels were made

1 Code
decides what it can: range hit, citation validity

→

2 Three LLM judges
from different model families, one rubric

→

3 AI adjudicator
settles disagreements, writes its evidence

No item was labelled by a person. Procurement specialists validate the labels in the pilot.

Limits and next steps

What is not built, what the public data cannot show, and what a pilot would add

Not built yet

- A check that a cited passage supports its sentence
- Vector Search over every PAD and ICR
- Daily job defined, paused until the shadow phase

Data gaps

- Amendments, milestones and payments are not public (STEP)
- Awards only from FY2020; no CPIA
- Lessons from 300 of about 8,100 ICRs

What the pilot adds

- Shadow mode, 4 to 6 weeks: answers logged, not shown
- 2 or 3 hubs, 15 to 30 specialists; staff decide
- Specialists label 100+ packages and check the AI labels

Small samples: 20 agent packages, 30 + 10 reviewed briefs; one brief moves the holdout rate by about 10 points

Demo

A new package in, a cited planning brief out



Thank you. Questions welcome.

github.com/ShabadVaswani/worldbank-procurement-intelligence

Shabad Vaswani